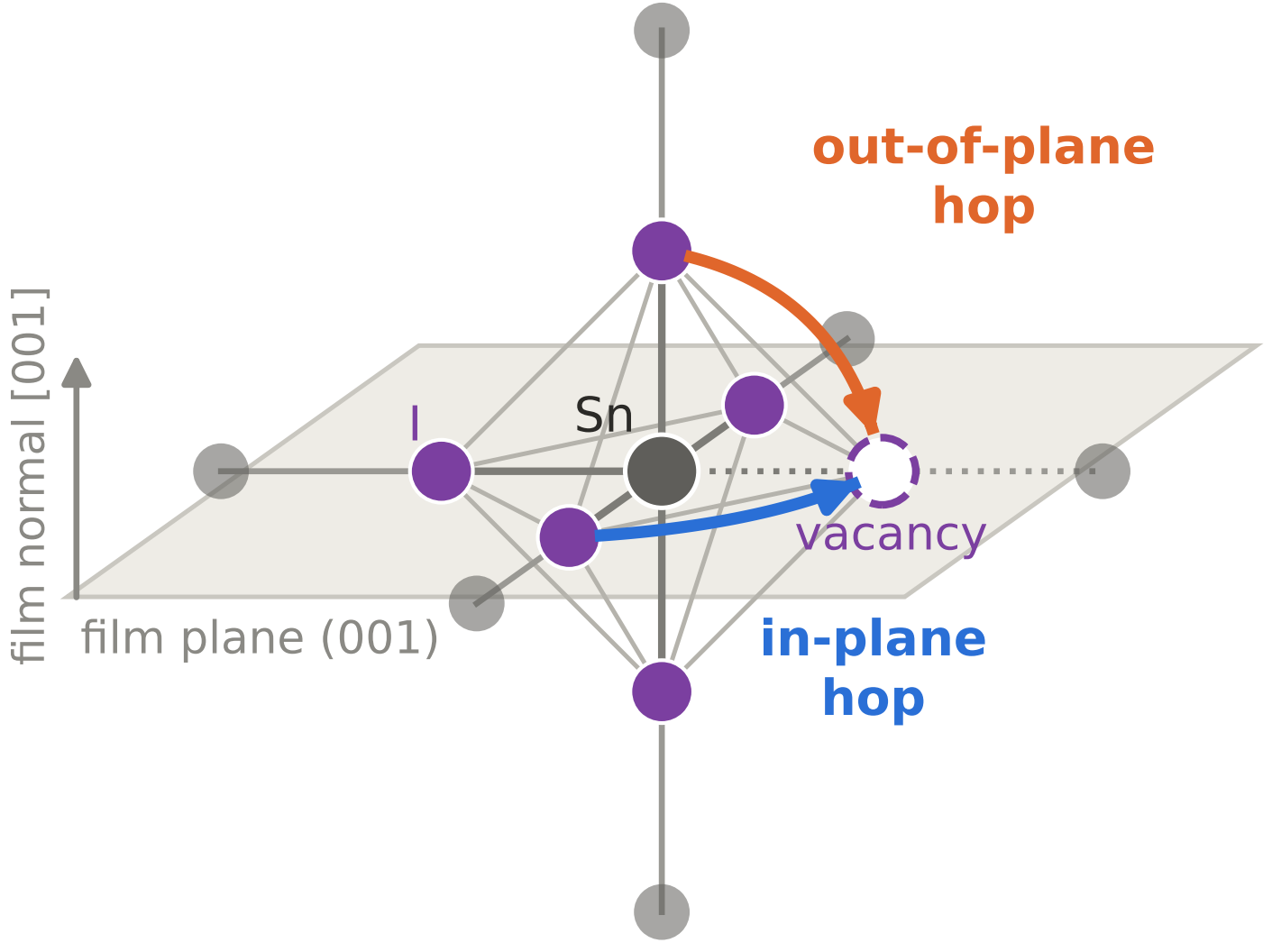
out-of-plane
hop
I
Sn
vacancy
in-plane
hop
film normal [001]
film plane (001)